

Azure Infrastructure Practice Questions & Answers

Practical Azure VM, Networking, Security, Storage and Troubleshooting Guide

Project context: Production Web Application Infrastructure & CI/CD Deployment using Azure Linux VM, VNet, Subnet, NSG, Azure Repos, Azure Pipelines and Nginx.

1. Resource Groups

Q. What is a Resource Group?

Answer: A Resource Group is a logical container in Azure used to organize and manage related resources. For example, it can contain a VM, VNet, NSG, Public IP, NIC and managed disks.

Q. Can one Azure resource belong to multiple Resource Groups?

Answer: No. An Azure resource can belong to only one Resource Group at a time.

Q. What happens when you delete a Resource Group?

Answer: The resources contained in the Resource Group are deleted, so the impact must be checked carefully before deletion.

2. Azure Virtual Machines

Q. What is an Azure Virtual Machine?

Answer: An Azure VM is a cloud-based virtual server. In this project, the Linux VM hosted Nginx, application files and required packages.

Q. What resources are commonly associated with a VM?

Answer: OS disk, optional data disks, NIC, VNet, subnet, NSG and optionally a Public IP.

Q. Difference between Stop and Deallocate?

Answer: Stopping stops the guest operating system. Deallocating releases the underlying compute resources; storage resources may still incur charges.

Q. How do you troubleshoot an unreachable Azure VM?

Answer: Check VM state, Public IP, NSG rules, required port, routing, credentials and SSH service. On Linux: `sudo systemctl status ssh`.

3. VNet and Subnets

Q. What is a VNet?

Answer: A Virtual Network is a logically isolated private network in Azure that allows resources to communicate securely.

Q. What is a subnet?

Answer: A subnet divides a VNet into smaller network segments, commonly separating web, application and database tiers.

Q. Can two VMs communicate without Public IPs?

Answer: Yes. They can communicate using private IP addresses when private network connectivity is configured.

Q. Scenario: Users should access the web server but not the database. How would you design it?

Answer: Place the web server in a web subnet and the database in a private subnet without a Public IP. Allow only required private traffic using NSG rules.

4. Network Security Groups

Q. What is an NSG?

Answer: An NSG contains rules that allow or deny inbound and outbound traffic based on source, destination, port and protocol.

Q. Which ports are commonly required?

Answer: 22/TCP for SSH administration, 80/TCP for HTTP and 443/TCP for HTTPS. Only required ports should be allowed.

Q. How should SSH be secured?

Answer: Restrict port 22 to trusted IP addresses, use strong authentication such as SSH keys where appropriate and follow least privilege.

Q. Difference between NSG and Azure Firewall?

Answer: NSG provides basic network traffic filtering at subnet or NIC level. Azure Firewall provides centralized and more advanced network security capabilities.

Q. Scenario: VM is running but SSH is not working. What will you check?

Answer: Check Public IP, NSG port 22 rule, SSH service, Linux firewall, username and authentication credentials or SSH key.

5. Public and Private IP

Q. What is a Public IP?

Answer: A Public IP enables internet-facing connectivity when associated with a resource and allowed by security rules.

Q. Should every VM have a Public IP?

Answer: No. Production workloads should use private connectivity where possible and expose only controlled entry points.

Q. Difference between Public and Private IP?

Answer: A Public IP can be internet reachable when allowed. A Private IP is used inside private networks and is not directly internet-routable.

6. Azure Disks and Storage

Q. Difference between OS Disk and Data Disk?

Answer: The OS disk contains the operating system and system files. Data disks are commonly used for application data, logs and backups.

Q. How do you check disk usage?

Answer: Use: `df -h`

Q. How do you find large directories?

Answer: Use: `sudo du -sh /*` and investigate specific locations such as `sudo du -sh /var/*`.

Q. Scenario: Disk is 95% full. What will you do?

Answer: Confirm the affected filesystem with `df -h`, identify large directories/files, safely remove unnecessary data, review log retention and extend storage if needed.

Q. How do you increase disk size in Azure?

Answer: Increase managed disk capacity in Azure, then extend the partition and filesystem inside Linux if required, and verify using `df -h`.

7. Monitoring and Performance

Q. Which metrics do you monitor?

Answer: CPU, memory, disk usage, disk I/O, network activity, VM availability and application/service health.

Q. How do you check memory?

Answer: Use: `free -h`

Q. How do you check CPU and processes?

Answer: Use: `top` or `htop`.

Q. How do you check Nginx status?

Answer: Use: `sudo systemctl status nginx`

Q. Scenario: CPU is above 90%. What will you do?

Answer: Identify the high CPU process, check logs and traffic, investigate the root cause, take corrective action and monitor after the fix. Do not restart blindly.

8. Website Troubleshooting

Q. Website is not accessible but Nginx is running. What will you check?

Answer: Test locally with `curl http://localhost`, verify listening ports, NSG rules for 80/443, Public IP/DNS, Linux firewall, Nginx configuration and application files.

Q. How do you check listening ports?

Answer: Use: `sudo ss -tulpn`

Q. Developer says the application is slow. Where do you start?

Answer: Check CPU, memory, disk I/O and network first, then application logs, web server behavior, traffic and database performance.

9. Production Security

Q. How would you secure a production VM?

Answer: Use least privilege, restrict administrative access, avoid unnecessary Public IP exposure, configure NSGs, patch regularly, disable unnecessary services, enable monitoring and alerts, configure backups and use HTTPS.

Q. Why is least privilege important?

Answer: It ensures users and services receive only the permissions they need, reducing security risk and accidental impact.

10. Project and CI/CD Practice Questions

Q. Explain your Azure infrastructure project.

Answer: I managed infrastructure supporting a web application on an Azure Linux VM. I worked with VM management, VNet, subnet, NSG rules, Nginx and controlled access. Azure Repos stored source code and Azure Pipelines automated deployment to the VM using SSH. I also handled monitoring, disk management and troubleshooting.

Q. Pipeline is successful but website is not updated. What will you do?

Answer: Verify the latest commit and branch, pipeline run, artifact/deployment output, deployed file on the VM, web server configuration and caching.

Q. How do you troubleshoot a failed deployment?

Answer: Start with pipeline logs, identify the failed task, then check source code, artifacts, service connection/SSH, VM availability, permissions, dependencies, disk space and deployment commands.

Q. What challenges did you face?

Answer: SSH authentication failure was resolved by checking service connection details, VM reachability and port 22. A missing unzip dependency was identified from logs, installed on the VM and the pipeline was rerun successfully.

11. Azure DevOps Practice Questions

Q. What is Azure Repos?

Answer: Azure Repos provides Git repositories for managing source code.

Q. What is Azure Pipelines?

Answer: Azure Pipelines automates building, testing and deploying applications.

Q. What is CI?

Answer: Continuous Integration is frequent integration of code changes with automated validation or builds.

Q. What is CD?

Answer: Continuous Delivery or Deployment automates moving validated changes toward or into environments.

Q. What is YAML?

Answer: A human-readable configuration format used to define pipeline triggers, stages, jobs and tasks as code.

Q. What is an artifact?

Answer: A pipeline output, such as a packaged application, used by later deployment stages.

Q. What is a service connection?

Answer: A secure connection configuration Azure DevOps uses to authenticate to external systems.

Q. How do triggers work?

Answer: Triggers automatically start a pipeline based on events such as a push to a configured branch.

Q. How do you roll back a failed deployment?

Answer: Keep previous stable versions/artifacts and redeploy the last known-good version after following the approved rollback procedure.

Final Project Summary

Flow: Developer → Azure Repos → Azure Pipeline → Build/Artifact → SSH Deployment → Azure Linux VM → /var/www/html → Nginx → Website

Practice tip: Focus on understanding the complete flow and explaining each component clearly. Use these questions to practice troubleshooting scenarios and practical concepts.